

Standardized Benchmarking of Multi-Agent Distributed Machine Learning in Augmented Reality

Research Update #1

Alexis Juarez Gomez & Parthkumar Joshi

1. Current State of Research

This overview looks at the overlap between three fast-moving fields: multi-agent systems (multiple AI agents working together), distributed machine learning (splitting ML workloads across devices instead of running everything on one server), and augmented reality (overlaying digital content onto the real world). Each area has matured on its own, but their intersection is still pretty new. As of now, no single widely adopted benchmark exists for testing multi-agent distributed ML specifically within AR environments.

1.1 Multi-Agent Systems & Benchmarking

Interest in multi-agent systems has exploded, and so have benchmarks. MultiAgentBench (2025) evaluates LLM-based agent coordination using milestone-based KPIs across different topologies (star, chain, tree, graph). GAIA and the Berkeley Function-Calling Leaderboard (BFCL) serve as independent yardsticks for agentic AI. However, none of this accounts for AR's real-time latency demands or spatial awareness requirements.

1.2 Distributed ML at the Edge for AR

AR applications require ultra-low latency (sub-20ms) for seamless overlay rendering, making edge computing essential. Research in distributed ML for AR has focused on model partitioning across edge-cloud ranges, where inference workloads are split between on-device, edge-server, and cloud resources. The scAtteR++ pipeline (deployed on the Oakestra edge-cloud framework) demonstrated distributed stream processing for AR but revealed significant challenges in network variability and service orchestration. Frameworks like TensorFlow Lite, ONNX Runtime, and Core ML enable on-device inference, while techniques such as quantization, pruning, and knowledge distillation compress models for resource-constrained AR headsets.

1.3 Federated Learning in XR

Federated learning (FL) trains models across devices without centralizing raw data, crucial for AR since these devices capture personal data like eye tracking and room layouts. A 2025 paper in npj Wireless Technology proposed Multi-Modal Multi-Task Federated Foundation Models (FedFMs) for XR and introduced the SHIFT framework covering five core challenges: Sensor diversity, Hardware assortment, Interactivity, Functional/task variability, and Temporality.

2. Ongoing Major Projects & Initiatives

Project / Framework	Focus Area	Key Contribution
FedFMs for XR (NSF-funded)	Federated foundation models for AR/VR/MR	SHIFT taxonomy; modular FedFM architecture for multi-modal XR tasks
FERMI-6G	Federated multi-agent RL for 6G edge networks	Cross-layer resource management with privacy-preserving MARL
scAtteR++ / Oakestra	Distributed AR stream processing at the edge	Practical edge-cloud deployment insights for AR pipelines
MultiAgentBench	LLM-based multi-agent evaluation	Milestone-based KPIs; coordination topology benchmarks

3. Key Challenges

- **No Unified Benchmark:** Current benchmarks tackle multi-agent coordination, edge ML, or AR performance separately. Nothing tests all three together under realistic end-to-end conditions with standardized metrics.
- **Real-Time Latency:** AR needs sub-20ms rendering loops. Distributing ML across multiple agents adds communication overhead and sync delays that are hard to predict and harder to benchmark consistently.
- **Hardware Assortment:** AR devices range from lightweight smart glasses with almost no compute to tethered headsets with full GPU access. A fair benchmark needs to work across this entire spectrum.
- **Multi-Modal Complexity:** AR workloads span vision, depth, audio, haptics, and spatial data simultaneously. Evaluating multi-agent systems that handle all of these in parallel requires metrics beyond simple accuracy.
- **Scalability & Credit Assignment:** Multi-agent environments face exponential joint action spaces and difficulty attributing results to individual agents, amplified in AR by spatial co-dependence on shared physical environments.
- **Reproducibility Gaps:** AR research relies on physical devices and environments that are hard to standardize. Existing simulators (AI2-THOR, Habitat) lack native support for multi-agent distributed ML or realistic network conditions.

4. Proposed Project: AR Flashcard Tutor

The AR Flashcard Tutor scans physical objects or printed cards via a device camera, then generates an AI-powered explanation, a quiz question, and a visual annotation overlaid on the live camera feed. Below is the recommended project specification.

4.1 Inputs

- **Primary:** Live camera feed (RGB, 720p/1080p, 30 fps) from a smartphone or tablet, capturing a printed flashcard or physical object.
- **Secondary:** User-selected subject domain (biology, chemistry, history, etc.) to contextualize AI output; optional difficulty toggle (beginner/intermediate/advanced).
- **Preprocessing:** Frame is run through object detection/recognition to identify the subject. OCR is applied if the input is a printed text card.

4.2 Outputs

1. **AI Explanation:** A 2–3 sentence explanation of the identified object/concept, displayed as an AR text overlay anchored to the card.
2. **Quiz Question:** A multiple-choice or short-answer question rendered as an interactive AR panel the user can tap to answer.
3. **Visual Annotation:** Labeled bounding boxes, arrows, or diagram overlays highlighting key features of the object (e.g., parts of a cell, regions of a map).

4.3 Datasets

Dataset	Purpose	Details
ImageNet (ILSVRC)	Object recognition backbone pre-training	14M+ images, 1,000 classes. Used as a pre-training base for transfer learning
COCO	Object detection and segmentation	330K images, 80 object categories with bounding boxes and segmentation masks
Open Images V7	Extended detection with visual relationships	9M images, 600 classes with relationship annotations useful for overlay generation
SQuAD 2.0 / TriviaQA	Quiz generation training/fine-tuning	QA pairs for evaluating and fine-tuning the text generation component
Custom Flashcard Dataset	Domain-specific card recognition	Self-collected 500–1,000 photographed flashcards across target subjects, annotated with labels and bounding boxes

4.4 Models

Component	Model	Rationale
Object Detection	YOLOv8-Nano or YOLOv9-Tiny	Real-time inference (<10ms on mobile), small footprint, strong accuracy-speed tradeoff for on-device AR
OCR	PaddleOCR or Google ML Kit (on-device)	Lightweight, multi-language, runs on-device for low-latency text extraction from cards
Explanation + Quiz Gen	GPT-4o-mini API or Phi-3-mini (on-device)	GPT-4o-mini for cloud-based generation; Phi-3-mini (3.8B params) for offline/edge use
Visual Annotation	SAM 2 (Segment Anything Model 2)	Zero-shot segmentation for precise overlays; use distilled version for mobile deployment

4.5 Simulator / Development Environment

Component	Recommendation
AR Platform	Unity 2022 LTS + AR Foundation (wraps ARCore/ARKit). Cross-platform AR rendering, plane detection, image tracking, and camera access.
Simulation / Testing	Unity XR Simulation Environment for device-free testing. Supplement with AR Foundation Remote for live device streaming during dev.
ML Integration	Unity Barracuda or Sentis for on-device ONNX inference (YOLO, segmentation). REST API calls to cloud LLM for explanation/quiz generation.
Alternative (Web)	WebXR + Three.js + TensorFlow.js for a browser-based prototype. Lower barrier to entry, runs on any modern smartphone browser, but limited AR tracking.

4.6 End-to-End Pipeline

Camera captures frame → YOLOv8-Nano detects/classifies object → PaddleOCR extracts text → class label + OCR text sent as prompt to GPT-4o-mini (or Phi-3-mini offline) → LLM returns explanation + quiz in structured JSON → SAM 2 generates segmentation masks for annotation → Unity AR Foundation renders all overlays anchored to the physical card/object in real time.